

Dae Hyun Lee

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EDUCATION

University of Washington <i>M.S., Data Science</i>	Mar 2027
University of Washington <i>B.A., Mathematics</i>	Mar 2025

TECHNICAL SKILLS

- **Languages:** Python, Java, R, SQL
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM
- **Techniques:** Deep Learning (3D CNN, ViT), Computer Vision, Rag, Gradient Boosting, Recommendation Systems, LLM Applications, Feature Engineering, CV & Leakage Control
- **Data & Cloud:** Databricks (Spark, Genie, Model Serving), Apache Spark, Pandas, Numpy, Google Cloud, ETL / Data Pipelines
- **MLOps & Tools:** Docker, Hugging Face, Weights & Biases, CI/CD, Git, Tableau, Plotly

Honors & Awards

Databricks - UW Hackathon -- First Place <i>TerraCast</i>	Feb 2025
• https://devpost.com/software/terracast	
Databricks Student Fellowship Program <i>Selected as 1 of 39 (top 0.8%) from over 5,000 applicants</i>	May 2026 - Present
NAIRR Pilot Allocation (NSF)	Aug 2026 - Present
<i>Awarded 5,000 GPU-hours on PSC Bridges-2, Hugging Face compute grants, and a Weights & Biases MLOps license</i>	

WORK EXPERIENCE

Roy Lab, University of Washington <i>Research Assistant</i>	Oct 2025 - Present
• Built a patient-level multimodal deep learning pipeline for precision medicine in Alzheimer's disease, integrating structural brain MRI across 18826 of ADNI, NACC, OASIS-3 subjects; benchmarked 3D CNN, ViT, ResNet, and ResNeXt for segmentation and classification, and CN/MCI/AD staging, raising accuracy from 55% to 70% • Modeled disease heterogeneity at the individual-patient level by extracting an imaging-derived cognitive reserve biomarker from cross-validated prediction residuals, using subject-wise splitting and class-balanced sampling to prevent leakage; reframed residuals as interpretable features for patient stratification	
SNU VLDB Lab, Seoul National University <i>Research Intern</i>	Jun 2024 - Sep 2024
• Designed a training-free, variance-based dimension selection method to accelerate embedding distance computation: profiled per-dimension distributions across the vector collection, z-scored every dimension, and retained only high-variance dimensions that drive pairwise distances, keeping the original axes unlike PCA • Cut OpenAI embeddings from 1,536 to 64 dimensions (96% reduction) while reproducing 95% of full-dimensional top-k nearest-neighbor results; integrated into an HNSW index and benchmarked recall and latency on 100K+ vectors, documented in internal technical reports	

PROJECTS

CarePilot GitHub <i>LLM Agent for Real-time Hospital Referral</i>	Jun 2026
• Built CarePilot, an AI chatbot that recommends the most suitable hospital based on the user's situation, using an end-to-end Databricks stack (Lakebase, Genie, Model Serving) • Designed a dynamic weighted scoring system that re-ranks hospitals in real time (e.g. distance weight increases with case severity) and provides driving routes via OSRM	
TerraCast Github <i>ML-based Crop Yield Forecasting on Databricks</i>	Feb 2026
• Designed and led end-to-end LightGBM optimization pipeline on Databricks using Spark and Python for county-level crop yield prediction; engineered 45+ features from NOAA weather and regional datasets, applied crop-specific modeling and yield detrending to isolate pure weather signal from long-term trend	
Smart Skincare Github <i>KNN Recommendation Engine</i>	Jan 2026
• Built ingredient-based recommendation engine over 4,500+ skincare products; scored products by matching ingredient profiles to user skin type, penalizing flagged ingredients using Paula's Choice dictionary	
Optimizing Clean Water Supply GitHub <i>Geospatial ML for Water Quality Prediction</i>	Nov 2025
• Designed and implemented a spatially aware ML pipeline for the EY 2026 Clean Water Supply Challenge, engineering remote-sensing and climate features (Landsat, TerraClimate, ERA5, HydroRIVERS) and using two-stage modeling for DRP to improve robustness under distribution shift	
WatchDawg Github <i>Seattle Crime Analytics Dashboard</i>	Sep 2025
• Built production analytics dashboard on 500K+ Seattle PD incidents (17+ years) with real-time filtering and geospatial visualization; engineered 11-stage data cleaning pipeline reducing dataset noise by 35%	